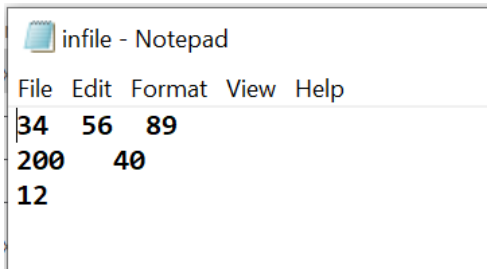


Lab 4

Objective: Practice Streams, Basic File I/O, Character I/O.

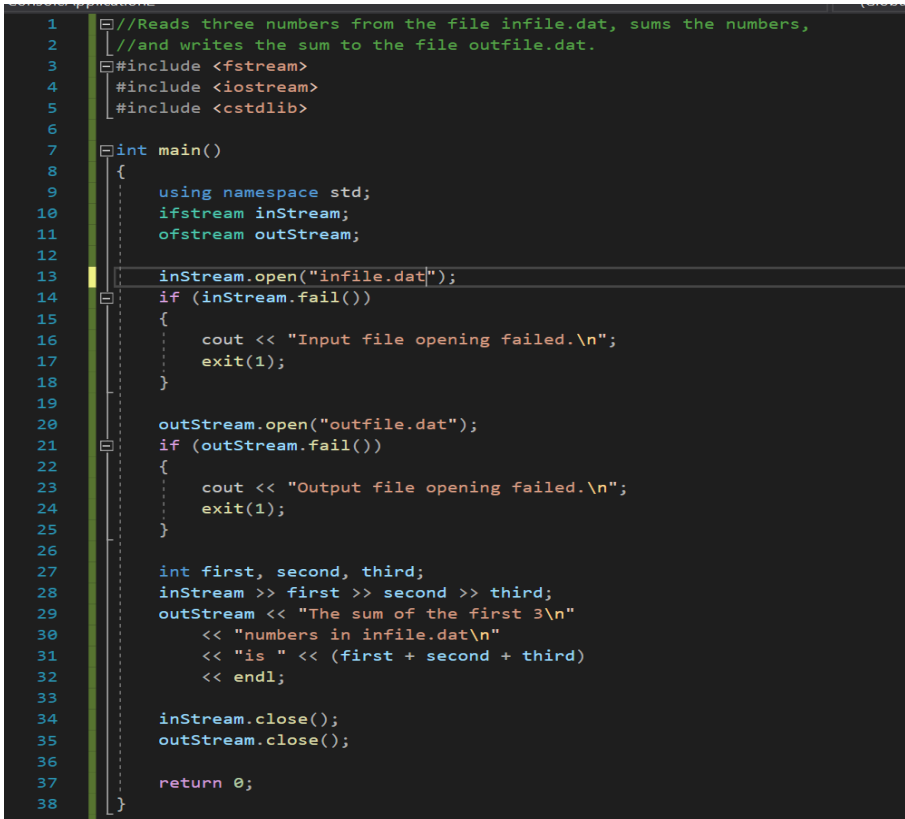
In this lab, you are going to practice I/O streams and character processing.

Task 1: practice simple file input/output by using the 6.1 sample code, you should create an input file called “infile.dat”, which looks like this:



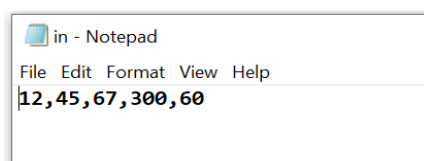
```
infile - Notepad
File Edit Format View Help
34 56 89
200 40
12
```

Finish the following code and run the program, check what’s in the output file called “outfile.dat”.



```
1 //Reads three numbers from the file infile.dat, sums the numbers,
2 //and writes the sum to the file outfile.dat.
3 #include <fstream>
4 #include <iostream>
5 #include <cstdlib>
6
7 int main()
8 {
9     using namespace std;
10    ifstream inStream;
11    ofstream outStream;
12
13    inStream.open("infile.dat");
14    if (inStream.fail())
15    {
16        cout << "Input file opening failed.\n";
17        exit(1);
18    }
19
20    outStream.open("outfile.dat");
21    if (outStream.fail())
22    {
23        cout << "Output file opening failed.\n";
24        exit(1);
25    }
26
27    int first, second, third;
28    inStream >> first >> second >> third;
29    outStream << "The sum of the first 3\n"
30        << "numbers in infile.dat\n"
31        << "is " << (first + second + third)
32        << endl;
33
34    inStream.close();
35    outStream.close();
36
37    return 0;
38 }
```

Task 2: practice character I/O, you should create an input file called “in.dat”, which looks like this:



```
in - Notepad
File Edit Format View Help
12,45,67,300,60
```

Finish the following code and run the program, check what's in the output file called "out.dat".

```
1  #include <iostream>
2  #include <fstream>
3  #include <cctype> // Include cctype for character classification
4  #include <string>
5  using namespace std;
6  int main() {
7
8      ifstream inStream;
9      ofstream outStream;
10
11     inStream.open("in.dat");
12     if (inStream.fail())
13     {
14         cout << "Input file opening failed.\n";
15         exit(1);
16     }
17
18     outStream.open("out.dat");
19     if (outStream.fail())
20     {
21         cout << "Output file opening failed.\n";
22         exit(1);
23     }
24
25     string number = ""; // To store each number as we extract them
26     int sum = 0;
27     char ch;
28
29     // Read character by character from the file
30     while (inStream.get(ch)) {
31         if (isdigit(ch)) { // If the character is a digit, add it to the number string
32             number += ch;
33         }
34         else if (ch == ',' || inStream.eof()) { // If a comma is encountered, process the number
35             if (!number.empty()) {
36                 sum += stoi(number); // Convert the string to an integer and add to sum
37                 outStream << number << "\n";
38                 number = ""; // Reset the number string for the next number
39             }
40         }
41     }
42
43     // Handle the last number if the file does not end with a comma
44     if (!number.empty()) {
45         outStream << number << "\n";
46         sum += stoi(number);
47     }
48     outStream << "The sum of the numbers are: " << sum;
49
50     // Close the files
51
52     inStream.close();
53     outStream.close();
54
55     return 0;
56 }
57
```

Finish the following code and run the program, check what's in the output file called "out.dat".

Submission:

You need to upload: the source code (.cpp files), your input text files and the output files on Moodle before 4pm Wednesday.